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Spatial 1:1 Models. A Tool for Collaborative and Participatory Processes in Housing Projects

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Abstract: Architectural production relies, among other methods and tools, on models, mock-ups and prototypes. This article proposes using spatial 1:1 models to communicate and design socio-spatial aspects in housing projects, thus, discussing the specificity of spatial 1:1 models and how they differ from material mock-ups or architectural prototypes. To demonstrate the relevance of spatial 1:1 models, a case study is presented. To illustrate the specific characteristics and possibilities of representation and interaction in the context of today's cooperative housing, the case study addresses threshold spaces and their social implications. Within dense cooperative housing projects these spatial configurations and the way they are communicated to future residents is critical for socially sustainable projects. The spatial 1:1 model investigates the acceptability and appropriability of threshold spaces and is intended to be an open platform for participation and collaboration. The article addresses the added value of a spatial 1:1 model for the design and communication process between lay people, architects, and various other stakeholders and its ability to improve socio-spatial qualities in the built projects.

Keywords: 1:1 Model; Mock-Up; Participatory Design Process; Cooperative Housing; Threshold Spaces.

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Introduction

Today's cooperative housing projects in a Central European context face socio-economic challenges that result in a shift in architectural typologies and adaptation in spatial configurations. The increase in construction and utility costs and the urge to build affordable housing results in more compact and dense projects. (European Commission 2019) It is not uncommon for detached, linear buildings to be replaced by larger volumes with narrower exterior spaces that result in densified neighborhoods and compressed spaces between neighbors.

Alternative life models and family structures cause smaller households and thus, smaller average apartment sizes (Bouzarovski et al. 2006). Together, these aspects lead to adapted circulation typologies and new forms of transitional spaces between the public and the private. This points to a crucial aspect of a successful housing project: the social acceptance and therefore the social sustainability of large housing developments. There are numerous failed social housing projects in architectural history; especially if architecture is pushed to rapidly adapt to socio-economic forces or experiment with new forms of living. Alison and Peter Smithson's ambitious and well-intended »streets in the sky« at Robin Hood Gardens, for example, pushed social density too far and did not secure acceptance by the residents which resulted in vandalism, mismanagement and ultimately the demolition of the project 45 years after its construction (Liakoura/Lozano 2020). To help avoid misplanning from a social perspective, a multitude of strategies can be applied. A participatory design process, thorough communication and interaction with future residents, is one of them.

As part of such a strategy, this article introduces the spatial 1:1 model which addresses the socially crucial threshold spaces in dense housing projects. Spaces can be tested and communicated before construction to reduce the discrepancy between projection and the built reality of a project. First, the specific benefits of spatial 1:1 models and their differentiation from architectural models and mock-ups are elaborated. The focus in the 1:1 scale is shifted from constructive material mock-ups toward social and spatial models for participation and collaboration. Second, the article examines a case study which was built during the summer of 2023 in the context of a cooperative housing project in Stuttgart. By recreating multiple extracts of the larger housing project and reconfiguring them in a spatial 1:1 model, future residents, architects, and other stakeholders could experience the

spaces, interact with them and test different settings of everyday life thereby overcoming some of the challenging abstractions of scaled models, plans or visualizations. The case study allows us to evaluate the general use of such models and their potential effect on architectural production, potentially leading to more socially sustainable housing projects and less social misplanning in the future.

Models, Mock-Ups & Prototypes

The terms model, mock-up, and prototype have different meanings and applications throughout different professional fields ranging from computer science, mathematics, philosophy, and psychology to linguistics and beyond. In this article, their meaning and use in the realm of architecture will be explored. Whereas the prototype is only sparsely used in today's architectural discussion, the model and mock-up are common terminologies, although boundaries remain fluent.

Accoding to the Oxford Dictionary, the prototype is often seen as a first or preliminary version of a device or vehicle from which other forms are developed. This implies the specific use of the term in architecture. A prototype as described above is the first version of a mass-produced product. In the realm of architecture, this occurs either with architectural elements such as doors, windows or other functional objects, prefabricated elements or with whole structures that are reproduced in large numbers. Some architectural movements in the past tended to focus more on the prototype as an architectural method. One was the Bauhaus when architects aspired to mass-produced products and architectural elements. Today, the term prototype is used in relation to single elements or products as well as in research, for example in the fields of robotic fabrication or experimental structural design.

Following the explanations of philosopher Herbert Stachowiak in his 1973 book *Allgemeine Modelltheorie*, the concept of a model can be defined on the basis of three main characteristics:

> »1. Mapping: Models are always models of something, i.e. mappings from, representations of natural or artificial originals, that can be models themselves. 2. Reduction: Models in general capture not all attributes of the original represented by them, but rather only those seeming relevant to their model creators and/or model users. 3. Pragmatism: Models are not uniquely assigned to their originals per se. They fulfil their replacement function a.) for

> particular-cognitive and/or acting, model-using subjects, b.) within particular time intervals and c.) restricted to particular mental or actual operations« (Stachowiak 1973: 131).

This applies for abstract models in other fields but especially to the physical and digital models in architectural practice. In contrast to the abstract concept of a model the architectural model, both physical and virtual, is usually utilized to represent a structure yet to be realized.

The architectural model often serves as a descriptive means of communication between laymen and experts to clarify spatial or constructive questions – thus becoming a collaborative and participatory tool. It appears to be particularly effective for this purpose as it allows for a rapid and holistic understanding of an architectural project and requires less imagination than drawn plans and sections (Gerkan 1994). In most cases, the model is related to a specific building project and is part of, and a tool in, the design process. Consequently, it is to be understood as an intermediate stage toward the eventual piece of work rather than as a body of work in its own right.

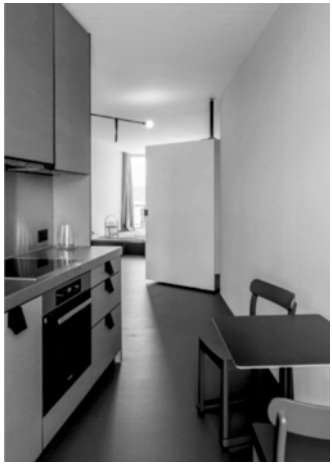
In most cases the architectural model is a scaled model, representing measurements and spaces with coherent relations but in an overall smaller scale. This requires a certain imagination to translate the perceived scaled model into a potential life-size project. Imagination and translation can be a challenge and might be overcome by introducing a model at a scale of 1:1. That said, what makes a 1:1 structure still to be a model? It is the properties mentioned previously. The 1:1 model is still an intermediate step toward a project. It emphasizes specific aspects of the project and depicts certain extracts that do not represent the whole. One type of 1:1 model is the mock-up. The mock-up is generally known to test materials and construction methods on a scale of 1:1 before construction starts. Besides typical constructive facade mock-ups, the search for more sustainable building materials and construction methods led to an increased use of the technical 1:1 mock-up. For example, the clay mock-ups for the project »Sitterwerk St. Gallen« by Roger Boltshauser (2017) developed in the context of the *École Polytechnique Fédérale de Lausanne* (EPFL Lausanne) (Boltshauser 2020). Such mock-ups lead to professional collaborations before the main project's construction begins and may also function to form trust and establish relationships between contractors and stakeholders. A collection of such examples and their value for architectural planning were the focus of the exhibition »Mock Up« at the *Swiss Architecture Museum* (SAM) in 2021. In the exhibition text,

curator Kofler describes the mock-up as a »true-to-material demonstration model that reproduces a section of a building at the original scale« (Kofler 2021: 2). Looking at the presentation in the exhibitions at SAM or the photo series »Archetypes of Contemporary Architectures« by David Ross or »The Mock Series« by Tobias Wootton, the mock-up itself, although originating as a experimental object in a larger design process, becomes the center of interest on its own. These further interpretations of the mock-ups, although interesting, are less relevant to the primary use of mock-ups or 1:1 models as discussed in this article. Although the 1:1 model has been used in architecture for quite some time and has proven itself as a valuable method, it is relatively unexplored today (Menz 2021).

In this context, the 1:1 scale, widely used in material mock-ups investigating construction techniques, will be reinterpreted here and used to communicate and explore spatial aspects of a project in an open and participatory process. Unlike the technical mock-up, the spatial 1:1 model can be understood by lay people and therefore enhance their participation and the integration of their opinions. Examples of the use of spatial 1:1 models can be found but are not well documented or publicly reflected on:

The »DesignBuild« project »Musterwohnung St. Quirin« was created in Munich in 2021 by the housing cooperative *Kooperative Großstadt eG* and the *Chair of Design and Construction* (by Prof. Florian Nagler) at the *TU Munich* to question contemporary forms of housing and their spatial structure in the floor plan (Nagler 2021) (fig. 1). The model was designed and constructed by students of the Chair at St.-Quirin-Platz in Munich and used as an open platform for a discussion with experts and city residents. Although a very interesting approach, the project was not directly linked to a specific housing project testing spatial aspects yet to be built. Instead, the project focused on a general discourse. Similar to the »Musterwohnung St. Quirin« is the concept behind Professor Ralf Pasel's summer academy »Werkstatt Multihalle« in Mannheim in the summer of 2019. The project was intended to test modular living spaces. To this end, students of the *TU Berlin* built prototype room modules in the *Multihalle Mannheim*, using a simple wooden construction method, over a period of two weeks and then set them up temporarily. Residents of the adjacent district were encouraged to participate through user workshops and were actively involved in the process. This project was mainly rooted in broader research about living spaces and was highly experimental. In both projects, the design and materiality were greatly simplified







1.
Chair of Architecture and Construction Florian Nagler, TU Munich: *Musterwohnung St. Quirin, 2021. Photographer: Matthias Kestel.*

2.
Chair for Architecture and Design Elli Mosayebi, ETH Zurich: *»Vacancy–no vacancy« 1:1 Housing Model, 2022. Photographer: Michael Stirnemann.*

3.
Summacumfemmer: *Mock-up of »Winter Garden« Facade for the San Riemo Housing Cooperative Building in Munich, 2022.*

and the spaces abstracted. The use of wood allowed for simple construction, with a screw and plug system that untrained students could assemble.

In contrast, the 1:1 housing model project »vacancy – no vacancy« (2019–2021) run by the *Chair of Architecture and Design* (Prof. Elli Mosayebi) at *ETH Zurich* was dedicated to question new concepts of living in the context of a real housing project before its construction. The 1:1 model was used to research the behavior of occupants in the interior spaces of the apartment- prior to the construction of idential apartments in the residential building Stampfenbachstrasse built by *EMI Architects* in Zurich (2018–2022). For this purpose, the behavior of the temporary residents (a few days or weeks at a time) in the 1:1 model was measured in regards to the use of performative interior fittings, such as a moving interior door. The project served as both a research apparatus and for the refinement of a concrete building project. Terminologically, the model was described by the authors as a »Mock-Up with prototypical elements« which again show the soft borders between the terms mock-up, model and prototype in the architectural context. Qualitative interviews and quantitative measurements with sensors on the interior furniture were used for evaluation (Mosayebi 2022) (fig. 2).

In the context of the cooperative housing project »San Riemo« (2017–2020), in Munich, the architecture office *Summacumfemmer* also created a 1:1 model, which they called »disobedient Mock-Up« or »Constructive Trojans«. The life-size model was argued to be a tool to manage the different opinions of contractors and site managers, norms and regulations during the design process when architects try to »transcend conventions«. Furthermore, Femmer and Summa state that

> »the architectural (near-) reality of the mock-up already enforces built facts, which are less contestable than prior drawn propositions. In this sense, mock-ups can make things more plausible, more convincing, but also more persuasive« (Summa/Femmer 2022).

They also argue that in their project, »San Riemo«, their proposals would have been less successfull if communicated through conventional means,

> »even scaled Models – would have fallen on death ears. […] And only the mock-up proved to speak a language that all of the participants could clearly hear, at the same time with a rhetoric that was convincing enough to gener-

> ate understanding and plausibility. The mock-up triggered a communicative proces« (Summa/Femmer 2022).

Thus it could be argued that for them, the mock-up played a central role in the realization of the prize-winning architectural project »San Riemo« although their »Constructive Trojans« are more of a mock-up than a spatial model they do have spatial and participatory qualities, therefore going beyond the pure constructive facade mock-up (fig. 3).

Although the differentiation between mock-up, 1:1 model, and prototype can be made linguistically, the examples show that the use of the terms and their intended meaning are not clearly separable. Nevertheless, the examples reveal a potential in research and practice on spatial 1:1 models and their use as collaborative and participatory tools, especially in the context of cooperative housing projects. The participatory aspect of spatial 1:1 models and the refinement of the project's spatial aspects bear potential for the outcome of the whole project.

Threshold Spaces

The »Schwellenraum« (threshold space) is linguistically composed of »Schwelle« (threshold) and »Raum« (space). It is therefore, a space but at the same time a transition between spaces. Only through the transition from one space to another does a person create the possibility of entering or leaving a space. Without a transition, there is no connection between the inner and outer spatiality. The creation of thresholds arises directly from the need to form space. (Boettger 2014: 10) In architecture, threshold spaces refer to the space between two spaces, between inside and outside, public and private, our body and what surrounds it. (Girot et al. 2021). Questions about the degree of visibility, accessibility, privacy, protection, blending and separating immediately arise in this context. In architecture, on one hand, these spaces can function as buffer zones in both climatic and socio-economic contexts. They are used, for example, as climatic layers to store heat or provide ventilation, or allow for common spaces while at the same time distancing private spaces. On the other hand, transitional spaces often serve purely functional purposes of circulation and access. The corridor, the foyer, or the vestibule often indicate the functional character of the threshold space, which sometimes has little to no quality as a place to stay due to economic and fire-protection motivations (Boettger 2014). In contrast to the functional

aspects, thresholds can also refer to the atmosphere of the spaces to which they allow access, thus providing a spatial and atmospheric transition.

In recent years, transitional spaces have once again gained importance in housing projects throughout Europe. The architects Lacaton and Vassal, for instance, have been dealing with the question of layering the facade for decades. In their »Cité du Grand Parc« project, a 2017 social housing project in Bordeaux, a new spatial layer in front of the existing building contributes to the summer extension of the apartments and thus to an increase in the quality of living for the residents. Similarly, the project »Cité manifeste« in Mulhouse (2005), opens the interior to the outside and turns it into a semi-private communal space. In particular, the shift toward smaller households and reduced flat sizes (in total square meters, not in square meters per person), as argued in the introduction, lead to the challenge of economically accessing many small apartment units. These circulations are increasingly designed as exterior circulation spaces like exterior corridors, access balconies, stairs or communal balconies. By doing so,the amount of insulated floor area can be reduced and consequently, utility costs are lowered. These factors led to a recent proliferation of access galleries in affordable housing. Apart from the economic advantages, prejudices against access galleries are still widespread among professionals and residents alike:

> »It should be said that access galleries the way we know them in conventional residential construction only meet with little approval among residents. They are perceived to be uninviting rather than inviting as recreational spaces. The walking areas are too narrow. The building fronts are closed above eye level to compensate for the narrow situation and the lack of distance to the apartments, which leads to windows that work as skylights. Social control is just as impossible in these cases as it is in closed interior hallways or on the stair landing in a hall/stairway structure. Seen exclusively as traffic carriers, access galleries are generally set along the climatically disadvantageous north side of a building. They, therefore, do not receive sunlight and are not adequate for the cultivation of plants. Hence it is understandable that residents prefer closed apartment access ways under these conditions« (Schneider 1995: 37).

For these reasons, architects try to design these often complex spaces in an inclusive manner: as common spaces which enable the development of a social milieu of a house community and increase the quality of living



4.
SSS Architecture & Research, Architecture, Studio Vlay Streeruwitz *and* EMT Architekten: *Urban plan of the cooperative housing project »Quartier am Rotweg«.*

5.
studio komaba: *Plans for the cooperative housing project »Quartier am Rotweg«. Transitional spaces between the private apartments and the common spaces, access balconies, etc.*

within the housing development. Sometimes, these generous shared spaces can compensate for smaller private spaces. At least that is the assumption and hope of many architects, developers, and housing cooperatives. The transition from inside to outside, private to collective becomes a key concern in these multi-functional threshold spaces. If constructed purely as described by Schneider, a lack of privacy and a high volume of transit can lead to anonymity and unwanted social friction. Therefore, a thorough and open design process for threshold spaces in housing projects seems to be crucially important for the social sustainability and overall success of a project. Consequently, the question of how to adequately represent and communicate such socio-spatial aspects like transitional spaces in a project become central. Boettger even states that these aspects are often neglected in planning due to aspects of economic efficiency and the lack of quantitative criteria (Boettger 2014). To counter the tendencies toward economic efficiency and functionality and focus more on social aspects, collaborative and participatory methods could be increased and greater attention paid to the design of threshold spaces. To do so, new methods should be developed for the participatory testing of the social aspects of housing projects and specifically of the threshold spaces. The spatial 1:1 model proposed here could be such a new method in a collaborative and participatory process to avoid social misplanning and strengthen increased socially aware architecture for the community.

Case Study: Spatial 1:1 Housing Model at Rotweg

On the urban periphery of Stuttgart, the housing project »Quartier am Rotweg« is to be realized by 2027. After the demolition of the site's existing post-war settlement, a lively, sustainable neighborhood with affordable housing and high-quality public spaces is to be built beginning in late 2023. Based on the winning project by *ISSS Architecture & Research*, nine dense residential buildings are planned. The central feature of each residential building is its external stair and corridor circulation combined with common spaces named »Vertical Commons«. These allow for efficient and low-cost access to a greater number of apartments. Reacting to the shift in household size and the intention to build affordable housing while densifying neighborhoods close to the core city. In conjunction with the planning of this cooperative housing project, a spatial 1:1 model was built during the summer of 2023. It was intended to test the socially important threshold spaces between communal





6.
studio komaba: *Extracts of the cooperative housing project and their inclusion into the spatial 1:1 plan.*

7.
studio komaba: *Photograph of the 1:1 model of one of the transitional spaces.*

and private living spaces and to investigate the use of a 1:1 model as a collaborative and participatory design tool. This accompanying project aimed to create a »lived culture of participation« through experimental working methods and participation activities. The organizing institutions were the housing cooperatives *Neues Heim* and *Zuffenhausen* in collaboration with the IBA'27 (International Building Exhibition). After an open call in 2022, *studio komaba* was selected to design and implement the 1:1 model in the summer of 2023 before the construction process for the larger housing development began in the fall of the same year. The spatial 1:1 model was used for appropriation, discussion, and participative processes that focus on testing threshold spaces and their social implications for the upcoming housing project. For this purpose, five transitions between the public and the private, the apartment and the circulation spaces were reconstructed. The model consisted of five different spatial situations as they are planned in the new larger housing project. Based on these situations, spatial and social aspects of the design could be openly discussed and also made tangible in life-size for non-professionals. Curated activities like workshops, discussions, or plays took place in the model. In one of the early workshops, potential future residents helped to furnish the spatial model and initiated a discussion about individual rooms, their allocated function, and their relation to the transitional spaces. Later in the process, various activities were organized by different actors under the guidance of the housing cooperatives.

Description of the Model

The 1:1 model combines five threshold spaces and spatial sequences as they are planned in the larger cooperative housing project. Threshold spaces, as well as parts of the private living spaces and common circulation spaces, were reconstructed. The model included two variations of »buffer zones« consisting of semi-heated winter gardens that separate the living spaces from the common circulation spaces with two layers of sliding doors. Furthermore, a situation was created whereby a bedroom was facing public access gallery. Here, the window size and proportion became crucial, as well as the distance between the access gallery and the interior room. In a fourth reconstruction, a living room was enlarged onto the common access gallery to create a semi-private outdoor seating area. Last but not least, the distance between two building parts and the courtyard they form was spatially reproduced by the model and could be experienced. This combination of different



8.
studio komaba: *Photographs of the 1:1 model showing different transitional spaces, materiality and performative elements.*

spatial situations forms a new whole, a model consisting of two parts that function as walk-through pavilions. The threshold spaces are tested not only horizontally, but also vertically, to simulate the three-dimensional situation in a multi-storey housing project.

As with all models, the question of abstraction, materiality, and extract became crucial for the spatial 1:1 housing model. The extracts are described above and derive from their contents. The appropriate materiality and degree of abstraction require specific strategies. As Gänshirt states in the context of architectural models, the interplay of building materials, design, and the statement of the model is certainly present, but not in a constraining relationship (Gänshirt 2012). Thus, not only forms or scales but also materials can be abstracted and represented by related (analogous) materials. This transformation in materiality is one of the central questions in planning a spatial 1:1 model because it is in contrast to the mock-ups not true in material or constructive methods. But rather, the spatial 1:1 model represents materials and spatial atmospheres in a coherent but translated manner.

Due to the model's size and temporality, it entails its own materials and construction methods which can differ from and need to be negotiated with the desired analogies with the larger project. Namely, the relationship between the model's construction materials, construction costs, and sustainability aspects become more important as the model's size increases. Despite these pragmatic motivations, the 1:1 construction method should still be able to support the overall claims of the model. After all, as Gerkan states: the construction method and the materiality of a model can often be used to reinforce the essence of the design or statement (Gerkan 1994). This is also true for spatial models in a 1:1 scale. The case study utilizes two material atmospheres: wood paneling and metal scaffolding. Both are suitable due to their ability to be deconstructed and the possibility of re-use. The planked wooden frame acts as an abstracted placeholder for plastered planar surfaces. Analogous to the grayboard at a scale of 1:100 or 1:50, wooden panels suggest a planar and opaque spatial boundary. The wooden paneling method was used in the 1:1 model for the interior »private« housing spaces–»the apartments«. Steel scaffolding, on the other hand, suggests a metallic, filigree construction method and was used in the model's communal outdoor spaces. The clear distinction between the different spaces and their materials accentuated the threshold between them and enhanced the perception of the transitions. The difference between the interior rooms and the exterior access balconies was therefore reinforced. For a more precise simulation of



9.
studio komaba: *Photographs of the 1:1 model showing different transitional spaces, materiality and performative elements.*

the transitions, additional performative elements such as vegetation, sliding doors, curtains, and furniture were incorporated. In order to simulate a more tangible, »inhabited« situation, which is crucial to the walk-through testing with lay people. It helped to simplify the understanding of the spaces and allowed easier access to, and interaction with, the model. Future residents, neighbors, or interested people were able to interact with the spaces, furniture, curtains, etc., and each other. In order to develop an understanding of how the threshold spaces may be used in the future. Performative components like furniture or curtains along with the changing vegetation made it possible to imagine and test different spatial configurations; not only for non-professionals but also for planners. The model increased the understanding of, and confidence in, the planning process.

Conclusion and Discussion

During the planning and construction process, as well as during the in-depth discussion about spatial 1:1 models as tools, the following aspects have become apparent. One is that the life-size model enables a deeper and broader examination of its specific topic; in this case, transitional threshold spaces in cooperative housing projects. The planning teams, site managers, contractors, and craftsmen have dealt intensively with the questions of threshold spaces in multi-story housing and how they can be formulated. Also, possible future residents, neighbors, and other stakeholders have been involved in the discussion and the understanding of the architectural challenges in building dense, cost-efficient housing has been increased. The relationship between flat size, circulation typology, costs, and ultimately, the form and dimension of the transitional spaces, has been communicated and lay people became more aware of crucial factors. Although difficult to quantify, it seems that through this participatory process and with the help of the 1:1 model, the acceptance and the willingness to adapt to even unconventional transitional spaces and the narrow borders between private and public spaces was increased. It remains to be seen whether and in which form this discussion will impact the design of the larger housing project to be built on-site. Naturally, it would be desirable if the 1:1 model was not only a representational tool for communication purposes but also an exploratory model that can feed new insights back into the ongoing building process. The 1:1 model will most definitely effect the »soft factors« of the design of the transitional spaces. By »soft factors« we mean the amount and type of vegetation, common and private furniture, per-

formative elements like sliding doors, curtains, or the choice of surfaces and materials. Even if these direct architectural implications may be limited, the spatial model can make a contribution beyond the housing project on-site. Evaluating the model's effects on the collaborative and participatory process – capturing the spatial experience of the visitors in the 1:1 model is a challenge and probably only possible with qualitative methods. Thus, the impact of the spatial 1:1 model will not be able to be measured precisely, even after the project is completed.

On a further note, the planning effort and costs of a 1:1 model cannot be neglected. The life-size spatial model is a costly approach which is not suitable for all housing projects or planning processes. Alternatives like immersive digital models or scaled models have their own disadvantages but may, in certain cases, be more effective. However, the effort and costs can certainly be justified for larger projects, especially if the 1:1 models are seen in a larger context. As Eidenbenz puts it, »economic profit is not suited for evaluating [constructive] mock-ups« and the same seems to be true for spatial 1:1 models. He states that the return of a constructive or material mock-up »is difficult, not measurable directly or becomes visible even only in later projects« And therefore

> »the decision of or against the use of mock-ups must not be linked to possible profit but must be made independently based on the desired quality, control and perceived responsibility to further develop the building culture« (Eidenbenz 2012: 210).

When comparing the spatial 1:1 model to the 1:1 material mock-up in particular, it can be stated that the design of project's socio-spatial aspects deserve just as much effort and financing as the constructive details do, since the social sustainability of housing projects has proven crucial for their longevity. So, the 1:1 model of the case study may on a larger time frame, contribute to the academic and architectural discourse for future projects, adding to the dialogue around cooperative housing projects.

In summary, the case study managed to improve the communication of socially relevant spaces in housing projects to a broader audience of stakeholders, contractors, and non-professionals, and therefore gives the topic of transitional spaces greater importance in the planning process. This can significantly improve the social aspects of a housing project and the community

of future residents, not only integrating them in participatory processes but also seeing them as future collaborators in the management and caretaking of, the designed spaces. On a more general note, spatial 1:1 models can support design decisions and the interdisciplinary discussion around them – and therefore contribute to an architectural discourse that focuses more on spatial qualities and social aspects.

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